

Physics Quest

Master project roadmap and asset documentation

Single source of truth for developers, artists, and AI agents.

Inventory date: 2026-08-17 | Engine: Godot 4.7 (GDScript)

Living status: docs/PROGRESS.md | Do not invent completed work.

PHYSICS QUEST — MASTER PROJECT DOCUMENT

Document type: Single source of truth for developers, artists, and AI agents

Project root: C:\Users\Mufrid Johanee\Desktop\capstone\final\

Godot project: final-godot/

Assets: asset/ (linked into Godot as res://asset/)

Last inventory date: 2026-08-17

Living status file: docs/PROGRESS.md (do not dump daily checklists into this master)

PDF export: docs/PHYSICS_QUEST_MASTER.pdf (regenerate after inventory changes)

Table of contents

1. Project overview
 2. Complete game structure
 3. Currently confirmed scenes
 4. Asset directory inventory
 5. Complete factory zone roadmap
 6. Factory asset master checklist
 7. Mini-game asset breakdown
 8. Other zones
 9. Zone status table
 10. Asset acquisition strategy
 11. Nano Banana generation standard
 12. Nano Banana prompt system
 13. Reference image rules
 14. File naming convention
 15. Asset cleanup pipeline
 16. Godot folder structure
 17. Audio asset plan
 18. UI master plan
 19. Current state summary
 20. Roadmap and dependencies
 21. Priority system
 22. Unknown / decision required
 23. PROGRESS.md rule
 24. AI developer rules
 25. Quick start for a new AI
 26. Evidence sources and how to use them
 27. Screenshot / visual confirmation notes
 28. Godot systems architecture
 29. Question-bank inventory
 30. Duplicate-file and import hygiene
 31. Gameplay-critical vs decorative assets
 32. Full factory scene script (docs vs Godot)
 33. Full mini-game design specs (authoritative GDD text)
 34. Complete Nano Banana prompt catalog
 35. Zone 02–05 documented designs (Map B)
 36. UI component master list
 37. Audio requirement matrices
 38. Character pose and animation matrix
 39. Godot import and sprite-scale rules
 40. Photopea cleanup procedure (step-by-step)
 41. What to implement next (recipes, not inventions)
 42. Scientist / flash-card battle (side concept only)
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How to read this document

Evidence is labeled. Do not upgrade a label without new proof.

Label	Meaning
CONFIRMED EXISTING	File or scene exists on disk in this repo
CONFIRMED IN DOCUMENTATION	Specified in GDD / proposal / chat dumps, not necessarily built
VISIBLE IN GAMEPLAY	Implemented and reachable in the Godot project
DOWNLOADED / AVAILABLE	File exists under <code>asset/</code> ; may be unused, duplicated, or unclear
GENERATED BUT NEEDS CLEANUP	AI/export file present; white bg, JPG, typo name, or not wired
PLANNED	Intended; design exists at some level
MISSING	Required for a documented feature; no usable file found
NOT YET DESIGNED	Mentioned as a zone/idea only; no floor-by-floor spec
UNKNOWN / REQUIRES DECISION	Contradictory or unspecified; do not invent

Hard rules

1. A filename in a design doc does **not** mean the asset is in Godot.
 2. A PNG on disk does **not** mean it is used in a scene.
 3. A placeholder `.tscn` does **not** mean the minigame is playable.
 4. Zones 02–05 are **not implemented**. Do not claim they are.
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1. PROJECT OVERVIEW

1.1 What Physics Quest is

Physics Quest is a **2.5D top-down educational RPG** in which a secondary-school student must prove they can *apply* physics, not only recall formulas.

The player begins in a **classroom**, receives an assignment from a teacher, then travels through **themed industrial / scientific sites**. At each site they face **dialogue, quizzes (typically $\geq 80\%$), and later hands-on challenges**. Completing a zone awards a **badge** and unlocks the next site on a **world map**.

CONFIRMED EXISTING engine: Godot 4.7.1, **GDScript**, project name `Physics Quest`.

CONFIRMED IN DOCUMENTATION (proposal): Godot 4 + **C#**.

UNKNOWN / REQUIRES DECISION: The capstone PDF listed C#. The running project is **GDScript**. Treat GDScript as the implemented stack unless the team explicitly migrates.

1.2 Core fantasy

The protagonist is a **back-bencher** who ignored class. The teacher gives limited time to complete physics assignments by visiting **real-world-flavored locations**, talking to experts (managers / director / later scientists), passing assessments, and solving factory emergencies.

Opening beat **CONFIRMED IN DOCUMENTATION** and **VISIBLE IN GAMEPLAY** (School cutscene):

- Teacher: *Welcome to Physics Quest! You need to secure at least 80% marks in your assignments to pass.*
- Student: *Understood, Sir! I will do my best.*
- Student stands, walks to the classroom door, fade to World Map.

1.3 Educational objective

CONFIRMED IN DOCUMENTATION

- Contextual physics (Bangladesh Class 9–10 syllabus appears in Zone 02–05 GDD).
- Assessment via quizzes and interactive challenges.
- Difficulty climbs with **Bloom's Taxonomy**: Remember → Understand → Apply → Analyze → Evaluate/Create.
- Formal evaluation ideas in the proposal: pre/post test, SUS, Hake's gain. **NOT implemented in code**.

VISIBLE IN GAMEPLAY today: Floor quizzes are 5 MCQs, pass $\geq 80\%$, JSON question files. These are **placeholder banks**, not the full 50-questions-per-floor GDD spec.

1.4 Intended technology stack

Tool	Role	Status
Godot 4.7	Engine	CONFIRMED EXISTING
GDScript	Game code	CONFIRMED EXISTING
C#	Listed in proposal	CONFIRMED IN DOCUMENTATION only
JSON question bank	Quizzes	CONFIRMED EXISTING (5 Qs × 4 floors)
Pixel art + sprite sheets	Characters / rooms	CONFIRMED EXISTING (partial)
Nano Banana / Gemini image gen	Custom assets	CONFIRMED IN DOCUMENTATION (workflow)
Photopea	BG remove, crop, PNG	CONFIRMED IN DOCUMENTATION
Piskel / Krita / Blender	Optional pixel / 3D	CONFIRMED IN DOCUMENTATION; unused in repo

1.5 Art direction (working)

CONFIRMED IN DOCUMENTATION (mixed):

- Original GDD / chat: **2.5D** top-down RPG, Pokémon Emerald–inspired pixel art.

- Zone 02 GDD: **2D** pixel, 1px dark outline, 2-tone cel, warm afternoon palette.

VISIBLE IN GAMEPLAY: Top-down classroom and factory floor paintings; character sheets are top-down / 3/4 views with pose labels.

UNKNOWN / REQUIRES DECISION: Officially "2.5D" vs "2D locked." Do not regenerate the whole game's art until this is decided. Match **existing classroom / factory floor / player sheet** when adding Zone 01 assets.

Primary factory **style reference** on disk:

- `asset/sprites/Environment/factory_asset_reference.png`
(**DOWNLOADED / AVAILABLE**; docs also call it `physics_quest_factory_asset_reference.png` — **filename mismatch**, see §22.)
-

2. COMPLETE GAME STRUCTURE

2.1 Conflicting zone maps (read this first)

Two documented maps exist. **Neither Zones 02–05 is in Godot.**

Map A — original chat / this roadmap prompt (CONFIRMED IN DOCUMENTATION):

Zone	Location	Discipline	Badge
01	Factory	Mechanics	Momentum Crest
02	Power Plant	Electricity	Spark Emblem
03	Research Lab	Magnetism	Flux Badge
04	Observatory	Optics	Lens Token
05	Industrial Facility	Thermodynamics	Heat Sigil

Map B — docs/physics_quest_zones_02-05_gdd (2).md (CONFIRMED IN DOCUMENTATION):

Zone	Location	Curriculum	Notes
01	Mechanics Factory	Measurement, Motion, Force	"locked, built" in that doc
02	Energy Refinery	Work, Power, Energy, Simple Machines	Full floor design + Foreman Rina
03	Matter Labs	Matter, Pressure, Density, Buoyancy	Skeleton
04	Wave Observatory	Heat, Waves, Sound	Skeleton
05	Light & Circuits Citadel	Optics, Electricity, Magnetism	Skeleton; possible 05a/05b split

On disk (DOWNLOADED / AVAILABLE, not wired as zones):

- powerplant_select.jpg
- researchlab_select.jpg
- world_map_background.png (used)
- No observatory or thermodynamics environment files found

UNKNOWN / REQUIRES DECISION: Which map is canonical. Until decided, **do not build Zone 02 scenes**. World map art may already imply Power Plant + Research Lab even if GDD Map B is preferred.

2.2 Master zone table

Zone	Location (Map A)	Discipline (Map A)	Badge (Map A)	Scene status	Evidence	Major gameplay	Asset status
01	Factory	Mechanics	Momentum Crest	Partially playable	Godot scenes School → Map → Exterior → F1–F4 → placeholder MGs	Dialogue, 80% MCQ, gating; MGs not real	Floors/player/teacher exist; Guard/Director unused; MG art loose files
02	Power Plant or Energy Refinery	Electricity or Work/Energy	Spark Emblem or unspecified	Not implemented	Select JPG on disk; GDD Map B written	PLANNED only	Select image only
03	Research Lab or Matter Labs	Magnetism or Matter	Flux Badge or unspecified	Not implemented	Select JPG on disk; GDD skeleton	PLANNED / NOT YET DESIGNED (depends on map)	Select image only
04	Observatory or Wave Observatory	Optics or Heat/Waves	Lens Token or unspecified	Not implemented	Docs only	NOT YET DESIGNED (skeleton in Map B)	MISSING
05	Industrial Facility or Light Citadel	Thermo or Optics+E&M	Heat Sigil or unspecified	Not implemented	Docs only	NOT YET DESIGNED	MISSING

2.3 Intended player journey (Zone 01)

CONFIRMED IN DOCUMENTATION (scene script in docs/capstone.md.txt):

1. School classroom assignment (black cutout / fade).

2. Appear in front of school, **walk to factory** with **road signs, player-controlled**.
3. **Guard** greets (they expected the student; assignment not turned in), **steps left**, door revealed.
4. Enter factory (fade).
5. Floors 1–3: manager blocks stairs until **80%** quiz.
6. Floor 4: Director quiz $\geq 80\%$ → emergency phone cutscene → **five minigames in one run** (any GAME OVER → restart MG1).
7. Award **Momentum Crest**, unlock Zone 02 on world map.

VISIBLE IN GAMEPLAY vs that script:

Beat	In Godot now?
School dialogue + walk to door + fade	Yes (automated cutscene, not free-walk from schoolyard)
Overworld walk school → factory with signs	No
Guard sprite + step-aside	No (ColorRect labeled "Guard")
Exterior uses <code>factory.png</code> cutaway	Yes (temporary)
Floors 1–4 quiz gating	Yes
Director emergency cutscene	No
Real 5 minigames + fail-to-MG1	No (placeholder ColorRect scenes chained)
Badge award + Zone 02 unlock	Flag exists; no UI / no Zone 02

3. CURRENTLY CONFIRMED SCENES (GODOT)

Playable chain **VISIBLE IN GAMEPLAY**:

```
School.tscn
→ WorldMap.tscn
  → Zone01_Exterior.tscn
    → Zone01_Floor1.tscn
      → Floor2 → Floor3 → Floor4
        → MiniGame01_Brake.tscn (placeholder)
          → MG2 → MG3 → MG4 → MG5 → WorldMap.tscn
```

Main launch scene: `res://scenes/School/School.tscn` (project.godot).

3.1 School / Opening — VISIBLE IN GAMEPLAY

Scene: `final-godot/scenes/School/School.tscn`

Script: `scripts/SchoolScene.gd`

Element	Status
Classroom background <code>classroom.png</code>	Used, full-viewport TextureRect
Teacher <code>teacher_idle.png</code>	Used
Student AtlasTexture on <code>player_transparent.png</code>	Used; seated <code>Rect2(1759, 848, 429, 607)</code>
Desks / blackboard / formulas / clock / globe / windows	Painted into <code>classroom.png</code> (not separate nodes)
Dialogue overlay (Teacher then Student)	Used (<code>DialoguePanel.tscn</code>)
Walk to top-left door, fade to World Map	Used (<code>SceneManager</code>)
"Next Scene" button	Hidden (legacy node)
Player WASD	Disabled during cutscene
Schoolyard / road to factory	MISSING as a scene
Audio (bell, chair, footsteps)	Files exist; not wired in <code>SchoolScene.gd</code>

3.2 World Map — VISIBLE IN GAMEPLAY (partial)

Scene: `scenes/WorldMap/WorldMap.tscn`

- Background: `world_map_background.png`
- One hotspot: transparent "Enter Factory" → Exterior
- **No** Power Plant / Lab / Observatory / Industrial buttons
- `powerplant_select.jpg` and `researchlab_select.jpg` are **on disk, unused**

3.3 Zone 01 Exterior — VISIBLE IN GAMEPLAY (placeholder)

Scene: `scenes/Zone01_Factory/Zone01_Exterior.tscn`

- Fill: opaque ColorRect + `factory.png` (F1–F4 **cutaway** graphic, not a true outdoor street)
- `factory_exterior (1).png` exists on disk, **not used**
- Guard: ColorRect body/head + label "Guard" — **not** `guard_shee (1).png`
- Button: Enter Factory → Floor 1

3.4 Zone 01 Floor 1 — VISIBLE IN GAMEPLAY

Scene: `Zone01_Floor1.tscn` + `Zone01Floor1.gd` + `ZoneFloorController.gd`

- Background: `factory_floor_1.png`
- Player WASD, collision **verified** (layer 1 vs walls layer 2)
- Dialogue → Start Quiz → ≥80% → Continue to Floor 2
- Empty `ActionBar` ColorRect **removed** on Floor 1 only
- Staircase lock area present

3.5 Zone 01 Floor 2 — VISIBLE IN GAMEPLAY

Scene: Zone01_Floor2.tscn

- Background: factory_floor_2.png (conveyor / arms / machinery **painted in the JPG/PNG**, not separate colliders for each machine)
- Same quiz/gating controller as other floors
- Empty ActionBar **still in this scene** (Floor 1 fix was Floor-1-only)
- Collision: runtime FloorRoomSetup walls, not Floor-1 baked machinery set

3.6 Zone 01 Floor 3 — VISIBLE IN GAMEPLAY

- Background: factory_floor_3.png
- Same quiz/gating pattern

3.7 Zone 01 Floor 4 — VISIBLE IN GAMEPLAY

- Background: factory_floor_4.jpg
- Quiz JSON: data/zone01_floor4_questions.json
- **No** Director sprite, **no** emergency cutscene, Continue currently goes toward minigame chain (placeholder)

3.8 Minigames 01–05 — PLACEHOLDER ONLY

Each MiniGame0X*.tscn uses PlaceholderScene.gd: ColorRect + title + Next.

Loose art (mg1_ramp_bg.png , crane.png , etc.) is **NOT** instantiated in those scenes.

3.9 Shared systems — VISIBLE IN GAMEPLAY

System	Files	Notes
Autoloads	GameManager, SceneManager, BadgeManager	Fade transitions; JSON save user://savegame.json
F5	GameManager	Resets flags; does not reload scene
Player	Player.tscn / Player.gd	Floors; School student is a separate Sprite2D
Quiz	QuizPanel, QuizManager	MCQ UI
Dialogue	DialoguePanel	One line + Continue
Verify	scripts/tools/verify_project.gd	Headless checks

4. ASSET DIRECTORY INVENTORY

Source: recursive listing of `asset/` on 2026-08-17.
Duplicates named `(2)` or `(1)` are Windows copies. **Do not use** `(2)` files (UID collisions). Canonical file = name **without** `(2)`.
Used In = referenced by a `.tscn` / `.gd` / `.tres` in `final-godot`. “None” means **DOWNLOADED / AVAILABLE but unused**.

4.1 Characters

Asset	Filename	Category	Current status	Used in	Needs cleanup?	Needs regen?	Notes
Player sheet (white bg)	<code>sprites/character/player.png</code>	Character	DOWNLOADED	No	Yes (opaque white)	No if transparent used	Source sheet
Player transparent	<code>player_transparent.png</code>	Character	CONFIRMED EXISTING	School, sprite frames	Edge speckle possible	Only if edges bad	2816×1536
Player frames	<code>player_sprite_frames.tres</code>	Character	CONFIRMED EXISTING	Player.tscn	No	No	Includes seated crop
Player walk strip	<code>sprites/animation/player_walking_animation.png</code>	Character	DOWNLOADED	None	Unknown	Unknown	Unused alternate
Teacher opaque	<code>teacher.png</code>	Character	DOWNLOADED	No	White bg	No if idle used	Source
Teacher transparent	<code>teacher_transparent.png</code>	Character	DOWNLOADED	No (School uses idle crop)	Speckle on shirt	Maybe	Full sheet
Teacher idle crop	<code>teacher_idle.png</code>	Character	CONFIRMED EXISTING	School.tscn	Minor speckle	Optional	429×587
Director sheet	<code>director_sheet.jpg</code>	Character	GENERATED BUT NEEDS CLEANUP	None	JPG, labels, no alpha	Crop + PNG	Not in Floor 4
Manager 1	<code>manager 1.jpg</code>	Character	GENERATED BUT NEEDS CLEANUP	None	Space in filename, JPG	Rename + PNG + crop	Not on floors
Manager 2	<code>manager 2.jpg</code>	Character	GENERATED BUT NEEDS CLEANUP	None	Same	Same	
Manager 3	<code>manager 3.jpg</code>	Character	GENERATED BUT NEEDS CLEANUP	None	Same	Same	
Guard sheet	<code>guard_shee (1).png</code>	Character	GENERATED BUT NEEDS CLEANUP	None	Typo name; Exterior uses ColorRect	Rename <code>guard_sheet.png</code> , crop	
Verify idle	<code>_verify_idle_on_purple.png</code>	Dev	DOWNLOADED	None	—	No	Debug image; do not ship

4.2 Environments

Asset	Filename	Category	Status	Used in	Cleanup	Regen	Notes
Classroom	<code>Environment/classroom.png</code>	Env	CONFIRMED EXISTING	School	No	No	Full room painting
Factory cutaway	<code>factory.png</code>	Env	CONFIRMED EXISTING	Exterior (temp)	Checker-like baked ground	Prefer true exterior	F1–F4 labeled building
Factory exterior	<code>factory_exterior (1).png</code>	Env	DOWNLOADED	None	Rename; verify alpha	If not matching style	Not wired

Asset	Filename	Category	Status	Used in	Cleanup	Regen	Notes
Floor 1	factory_floor_1.png	Env	CONFIRMED EXISTING	Floor1.tscn	No	No	
Floor 2	factory_floor_2.png	Env	CONFIRMED EXISTING	Floor2.tscn	No	No	
Floor 3	factory_floor_3.png	Env	CONFIRMED EXISTING	Floor3.tscn	No	No	
Floor 4	factory_floor_4.jpg	Env	CONFIRMED EXISTING	Floor4.tscn	JPG	Optional PNG	
World map	world_map_background.png	Env	CONFIRMED EXISTING	WorldMap.tscn	No	No	
Power plant select	powerplant_select.jpg	Env	DOWNLOADED	None	JPG	If used as zone art	Map A evidence
Research lab select	researchlab_select.jpg	Env	DOWNLOADED	None	JPG	Same	
Style reference	factory_asset_reference.png	Ref	DOWNLOADED	Docs/prompts	Not a gameplay sprite	No	See §11
Observatory	—	Env	MISSING	—	—	Yes when designed	
Thermo / industrial	—	Env	MISSING	—	—	Yes when designed	
School exterior / road	—	Env	MISSING	—	—	Yes for GDD scene 2	
Director office (unique)	—	Env	MISSING	Floor 4 uses floor_4.jpg	—	Optional	

4.3 Machinery / minigame loose files

These are **DOWNLOADED / AVAILABLE**. **None** are used in minigame `.tscn` files. Treat as **GENERATED BUT NEEDS CLEANUP** until sliced, named, and wired.

Asset	Filename	Likely MG	Used in	Notes
Crane	crane.png	MG3	None	
Press	press.png	MG4	None	
Rig	rig.png	MG5	None	
Pulley (typo)	pully (1).png	MG5	None	Rename <code>mg5_pulley.png</code>
Load	load (1).png	MG3/5	None	
Weights	weights (1).png	MG3	None	May be a sheet of 5 weights
Gauge (typo)	guage (2).png	MG4	None	Canonical should be <code>mg4_pressure_gauge.png</code>
MG1 block	mg1_block (1).png	MG1	None	
MG1 ramp bg	mg1_ramp_bg.png	MG1	None	Closest to a minigame background
MG2 gear	mg2_gear_a (1).png	MG2	None	Need more gears
MG2 housing	mg2_housing.png	MG2	None	
Badge	badge_momentum_crest (1).png	Reward	None	Not shown in UI
Conveyor (separate)	—	Env/MG2	MISSING	Floor 2 has conveyor painted in BG only

4.4 Audio (files exist; playback not confirmed in scenes)

File	Status	Used in code?
music/amb_classroom.ogg	DOWNLOADED	No (grep: not in SchoolScene)
music_opening_sting.ogg	DOWNLOADED	No
sfx_chair_scrape.ogg	DOWNLOADED	No
sfx_footsteps_wood.ogg	DOWNLOADED	No

File	Status	Used in code?
sfx_paper_rustle.ogg	DOWNLOADED	No
sfx_school_bell.ogg	DOWNLOADED	No
sfx_typewriter_tick.ogg	DOWNLOADED	No
Factory SFX set	MISSING	—

4.5 Fonts / icons

Item	Status
asset/fonts/	Empty / MISSING
Dedicated icon set	MISSING (Godot default theme used)

4.6 Data

File	Status	Notes
final-godot/data/zone01_floor1_questions.json ... floor4	CONFIRMED EXISTING	5 questions each; not 50/floor GDD
Save	user://savegame.json	Runtime

5. COMPLETE FACTORY ZONE ROADMAP

5.1 Structure (CONFIRMED IN DOCUMENTATION)

- Multi-floor factory. Chat: **5-story** building; playable spec: **Floors 1–4** + optional rooftop/Floor 5 establishing shot.
- Floors 1–3: Floor Manager + quiz, Bloom difficulty up.
- Floor 4: Director + final quiz (Analyze/Evaluate).
- Then emergency + **five minigames, one continuous run**.
- Success: Momentum Crest; World Map unlocks Zone 02.

5.2 Bloom mapping (CONFIRMED IN DOCUMENTATION; not encoded in JSON)

Floor	Bloom (docs)	In code
1	Remember → Understand	5 generic MCQs
2	Understand → Apply	5 generic MCQs
3	Apply → Analyze	5 generic MCQs
4	Analyze → Evaluate	5 generic MCQs

5.3 Minigames (authoritative design: docs/capstone,md.txt)

#	Name	Physics	Fail condition (docs)	Godot
1	Emergency Brake	$f = \mu mg$, N2, deceleration	Crash → GAME OVER → MG1	Placeholder
2	Gear Swap	Gear ratio, RPM	Wrong confirm → belt snap	Placeholder
3	Counterweight Balance	Moments $F_1d_1 = F_2d_2$	Wrong/timeout → crash	Placeholder
4	Pressure Panic	$P = F/A$, Pascal	Wrong zone → burst	Placeholder
5	Pulley Rush	$MA = \text{Load}/\text{Effort}$	$MA < 4 \rightarrow \text{snap}$	Placeholder

MG1 numbers (docs): mass 40 kg, $\mu = 0.35$, 7 m; YES/NO then extra force.

MG3 numbers: 300 N at 2 m, CW at 3 m; weights 150/180/200/220/250 N.

MG4 numbers: A_{in} 0.02 m², A_{out} 0.10 m², P_{out} 5000 Pa; forces 60/100/140 N.

MG5 numbers: Load 600 N, motor 150 N, need $MA \geq 4$.

Any GAME OVER **restarts Mini-Game 1. NOT IMPLEMENTED.**

5.4 NPCs (docs vs code)

NPC	Docs	Code
Teacher	School	School sprite
Guard	Exterior step-aside	Placeholder box
Manager 1–3	Floors 1–3	Dialogue text only; JPG sheets unused
Director	Floor 4 + MG5 clap	Dialogue-as-manager pattern; JPG unused
Workers	Optional hints	Missing

6. FACTORY ASSET MASTER CHECKLIST

Legend: [x] file on disk · [~] file but unused/unclean · [] missing · [G] Godot default UI

A. Characters

- [x] Player idle/walk (transparent sheet + SpriteFrames)
- [] Player talk pose (distinct)
- [] Player climb stairs
- [x] Teacher idle crop
- [~] Director sheet (JPG, unused)
- [] Director worried / smiling variants (not confirmed as separate files)
- [~] Manager 1/2/3 JPG unused
- [~] Guard sheet unused / Exterior placeholder
- [] Factory worker male/female
- [] Engineer / technician / safety officer (**PLANNED** in chat; not required for MVP)

B. Environment

- [x] Classroom
- [~] Factory cutaway used as exterior
- [~] factory_exterior unused
- [x] Floors 1–4 backgrounds
- [] School exterior + road + signs
- [] Door closed/open states
- [] Staircase locked/unlocked overlay (logic exists; art optional)
- [] Rooftop / Floor 5
- [x] World map painting
- [] Unique Director office if Floor 4 art is insufficient

C. Machinery (world props)

- [~] Conveyor only as pixels on floor_2.png
- [] conveyor_belt.png / broken
- [] crates, barrels, pressure plate, industrial button/switch
- [] pipes/valves as separate sprites
- [] control panel sprite
- [x] factory_asset_reference.png (style only)

D. Mini-game assets

See §7. Many raw files [~], none wired.

E. UI

- [G] Godot Button / Label / PanelContainer
- [x] Dialogue panel scene
- [x] Quiz panel scene
- [] Custom MCQ art, nameplates, portraits
- [] World map lock icons for zones 2–5
- [~] Badge PNG unused
- [] Pause, retry, GAME OVER, PROBLEM SOLVED banners

F. VFX

- [] Sparks, dust, steam, alarm overlay, confetti
- [] Some FX may exist **inside** factory_asset_reference.png as drawings, **not** sliced sprites

G. Audio

- ☒ 7 school/opening files (unwired)
- ☐ All factory / MG / quiz / badge SFX

H. Fonts

- ☐ Project fonts (Bengali+English if bilingual — **UNKNOWN**)

I. Icons

- ☐ Padlock, check, cross, interact, push/pull

J. Animation

- ☒ Player 4-dir via AtlasTexture frames (mostly single-frame “walk”)
 - ☐ Guard sidestep
 - ☐ Door open
 - ☐ MG success/fail animations
-

7. MINI-GAME ASSET BREAKDOWN

Existing? = file found under `asset/` (any name). **In scene?** = used by MiniGame tscn. All five: **In scene? No.**

MG1 — Emergency Brake

Asset	Purpose	Existing?	Download?	Nano Banana?	Animation?
Ramp / side-view bg	Playfield	Yes <code>mg1_ramp_bg.png</code>	No	Only if style mismatch	Optional scroll
Sliding block	Actor	Yes <code>mg1_block (1).png</code>	No	Cleanup	Slide + stop/crash
Data panel	mass, μ , distance	No (use Godot UI)	No	Optional	—
YES/NO buttons	Friction-alone question	Godot Button	No	No	—
3 force buttons	Extra brake	Godot	No	No	—
Speed indicator	Tension	No	Optional	Optional	Needle/number
Countdown bar	Timer	Godot ProgressBar	No	No	—
Screech / dust	Success	No	Optional	VFX sheet	Yes
Crash + sparks	Fail	No	Optional	Yes	Yes
PROBLEM SOLVED banner	Success	No	No	Optional	—
GAME OVER	Fail	No	No	Optional	—

MG2 — Gear Swap

Asset	Purpose	Existing?	NB?	Anim?
Housing + smoke	Broken slot	<code>mg2_housing.png</code>	Cleanup	Smoke loop
Gear A	Draggable	<code>mg2_gear_a (1).png</code>	Need 3–4 more sizes	Spin
Empty slot highlight	Target	Missing	Yes or Godot ColorRect	Pulse
RPM meter	Feedback	Missing (<code>guage</code> may be reused)	Maybe	Needle
Conveyor	Result	Missing as sprite	Yes	Belt move
Test / Confirm buttons	UI	Godot	No	—
Belt snap	Fail	Missing	Yes	Yes

MG3 — Counterweight Balance

Asset	Purpose	Existing?	NB?	Anim?
Crane arm	Tilting	<code>crane.png</code>	Cleanup	Tilt
Conical load	Left mass	<code>load (1).png</code>	Cleanup	Descend
Weight blocks 5×	150–250 N	<code>weights (1).png</code> (sheet?)	Slice + labels	Place
Shelf	Source of weights	Missing	Optional	—
Place button	Commit	Godot	No	—
Descent bar	Timer	Godot	No	—
Impact destroy	Fail	Missing	Yes	Yes

MG4 — Pressure Panic

Asset	Purpose	Existing?	NB?	Anim?
Hydraulic press	Two pistons + pipe	<code>press.png</code>	Cleanup	Needle/steam
Gauge 3 zones	Green/yellow/red	<code>guage (2).png</code>	Rename + verify	Needle sweep
Force buttons	60/100/140 N	Godot	No	—
Steam success	Win	Missing	Yes	Yes

Asset	Purpose	Existing?	NB?	Anim?
Pipe burst	Fail	Missing	Yes	Yes

MG5 — Pulley Rush

Asset	Purpose	Existing?	NB?	Anim?
Empty rig	Playfield	rig.png	Cleanup	—
Pulley parts	Drag-drop	puller (1).png (maybe one)	Need fixed/movable/double/spool	Thread rope
Engine block	Load	Maybe load	Distinct from MG3	Rise/drop
Motor	Effort 150 N	Missing (in reference sheet?)	Yes	Whir
MA readout	Live	Godot Label	No	—
Rope snap	Fail	Missing	Yes	Slow-mo
Director clap	Win beat	director_sheet unused	Crop	Optional

8. OTHER ZONES

Until §22 zone-map decision is made, treat Map A names as **prompt labels** and Map B as **alternate GDD**.

Zone 02 — Power Plant (Map A) / Energy Refinery (Map B)

- Physics / reward: **CONFLICTING DOCS** (Electricity + Spark Emblem vs Work/Power/Energy + Foreman Rina).
- Scene confirmed? **No**
- Playable? **No**
- Background available? **Partial** — `powerplant_select.jpg` only (Map A flavor)
- Map B has **full floor design** (Loading Dock, Drop Tower, Machine Shop, Control Room) in `physics_quest_zone02_scene_design (2).md` — **CONFIRMED IN DOCUMENTATION**, not built
- **Do not implement floors until the map is locked**

Proposed (NOT FINAL) if Map A wins: exterior, generator, turbine, control, quiz, badge.

Proposed (NOT FINAL) if Map B wins: follow that scene-design doc.

Zone 03 — Research Lab / Matter Labs

- `researchlab_select.jpg` on disk, unused
- Map B skeleton: states of matter, $P=F/A$, density, buoyancy
- **NOT YET DESIGNED** at Factory-level detail if Map A magnetism lab is chosen

Zone 04 — Observatory / Wave Observatory

- **No files found**
- Map B skeleton only
- **NOT YET DESIGNED**

Zone 05 — Industrial / Citadel

- **No files found**
- Map B flags possible split 05a/05b
- **NOT YET DESIGNED**

Side concept (chat dump) — scientist Pokémon battles + flash cards

- **PLANNED / OPTIONAL / NOT IN CODE**
 - Conflicts with current Floor = Manager quiz
 - **Do not implement** unless explicitly scheduled; would be a large second system
-

9. ZONE STATUS TABLE

Zone	Concept	Scene confirmed?	Playable?	Background available?	Characters available?	Gameplay designed?	Assets complete?
01 Factory	Mechanics	● School, map, exterior, F1-4	● Quizzes yes; MG no	● Floors yes; true exterior/guard no	● Player/teacher yes; NPC sheets unused	● Docs yes	● No
02	● Two competing concepts	●	●	● select JPG only	●	● Map B detailed / Map A named	●
03	● Competing	●	●	● select JPG	●	● Skeleton or unnamed	●
04	● Competing	●	●	●	●	● Skeleton	●
05	● Competing	●	●	●	●	● Skeleton	●

● Confirmed/ready · ● Partial · ● Planned · ● Missing · ● Not yet designed / conflict

10. ASSET ACQUISITION STRATEGY

Method	Use when
A — Existing project asset	File already in <code>asset/</code> and style-matches (classroom, floors, player_transparent)
B — Download	Generic UI/SFX with a clear license; not unique NPCs
C — Nano Banana	Custom pixel props/NPCs matching reference
D — Manual edit	JPG→PNG, Photopea alpha, crop sheets, rename, slice weights/gears
E — Programmer UI	Buttons, bars, labels, quiz layout (already mostly this)

Priority order for Zone 01: D (wire and clean what you have) → C (only true gaps) → E (UI) → B (SFX if needed) → never regenerate player/classroom without a reason.

11. NANO BANANA GENERATION STANDARD

Match:

- Retro pixel art, Pokémon Emerald-like readability
- Industrial / school science mood
- Muted purple-gray metal, dark outlines, 2-tone shade
- **No photorealism, no 3D renders, no UI chrome, no text** unless asked
- Transparent PNG, one object, centered
- Same “camera” as factory floors / character sheets (top-down or 3/4 as appropriate)

Upload as style reference for factory/mechanical objects:

`asset/sprites/Environment/factory_asset_reference.png`

This is a **STYLE REFERENCE**, not a drop-in spritesheet. Slice later if needed.

Do not use classroom.png as the style ref for a gear.

Do not use player.png as the style ref for a turbine.

12. NANO BANANA PROMPT SYSTEM

12.1 Individual asset template

Create a game-ready pixel-art [ASSET NAME] for Physics Quest, a retro 2.5D educational physics RPG.
Use the uploaded `factory_asset_reference.png` as the primary visual style reference.
Match its pixel density, dark outlines, muted industrial purple-gray palette, shading, proportions, and mechanical design language.
Create only the requested object, centered, isolated, transparent background, no text, no UI, no extra objects, no background scene.

12.2 High-priority factory / MG prompts (P0–P1)

Reference to upload: `factory_asset_reference.png` unless noted.

factory_conveyor_belt.png

Template with [ASSET NAME] = `side-view industrial conveyor belt segment, flat top, visible rollers, Pokémon Emerald industrial pixel style.`

factory_conveyor_belt_broken.png

Same belt, snapped belt, slight smoke, still isolated.

factory_movable_crate.png

`small wooden crate, 3/4 top-down, readable silhouette, no labels.`

factory_heavy_crate.png

`larger darker crate, metal corners, heavier read.`

factory_pressure_plate.png

`floor pressure plate, recessed metal, top-down.`

factory_switch_industrial.png

`wall lever switch, chunky pixel, isolated.`

factory_spring_mechanism.png

`metal coil spring standing upright, isolated.`

factory_fulcrum.png

`triangular fulcrum block for a lever, isolated.`

mg5_pulley_double.png

`double movable pulley block with two wheels, isolated, no rope text.`

mg3_counterweight.png

`single hanging counterweight, industrial, isolated.`

factory_gear_assembly.png

`two meshed gears on a small mount, isolated.`

factory_control_panel.png

`chunky factory control panel with blank gauges (no readable numbers), isolated.`

npc_factory_worker.png (also upload `player_transparent.png` for scale/style of characters)

`one factory worker, same top-down proportions as the uploaded player sheet, idle front pose only, isolated, no pose labels, transparent background.`

guard_sheet.png — only if `guard_shee (1).png` is unusable

Upload `player_transparent.png` + `factory_asset_reference.png`

`guard character sprite sheet matching player sheet layout (idle front, side, back), factory uniform, no photorealism. Prefer cleanup of existing guard file first (Method D).`

12.3 Other-zone prompts (P3 until zone map is locked)

Mark outputs as **PLANNED**. Upload zone select JPGs if generating matching thumbnails.

powerplant_generator.png

Upload `powerplant_select.jpg` + `factory_asset_reference.png`
`pixel-art electrical generator, isolated, no text.`

powerplant_turbine.png

Same refs: `industrial turbine, isolated.`

researchlab_magnet.png

Upload `researchlab_select.jpg`: `horseshoe or lab magnet, isolated pixel art.`

researchlab_coil.png

`electromagnet coil, isolated.`

observatory_telescope.png

No zone file — upload factory ref only and expect style drift: `pixel telescope, isolated.` **Better to wait for observatory concept art.**

observatory_lens.png

`convex lens icon/object, isolated, no text.`

industrial_heat_chamber.png / industrial_thermometer.png

Wait for Zone 05 decision; do not mass-generate.

13. REFERENCE IMAGE RULES

Generating	Upload
Gears, belts, pulleys, levers, doors, FX	factory_asset_reference.png
Crates, plates, worker props on factory floor	factory_asset_reference.png + optional factory_floor_1.png for palette
Player-like body / new student poses	player_transparent.png
Teacher variants	teacher.png or teacher_idle.png
Guard / manager / director cleanup	That character's existing sheet, not the factory prop sheet
Power plant thumbnails	powerplant_select.jpg
Lab thumbnails	researchlab_select.jpg
School furniture	classroom.png

Never upload a minigame ramp as the reference for a character.

14. FILE NAMING CONVENTION

- `snake_case`, English, no spaces (`manager 1.jpg` → `manager_1_sheet.png`)
- Prefix by domain: `factory_`, `mg1_`, `mg2_`, `school_`, `ui_`, `sfx_`, `amb_`
- No (1) / (2) copies in the live set
- Fix typos when renaming: `pully` → `pulley`, `guage` → `gauge`, `guard_shee` → `guard_sheet`

Examples:

```
factory_asset_reference.png
factory_conveyor_belt.png
factory_movable_crate.png
mg1_emergency_brake_block.png
mg1_ramp.png
mg2_gear_small.png
mg2_gear_housing.png
mg3_crane_arm.png
mg3_weight_150n.png
mg4_hydraulic_press.png
mg4_pressure_gauge.png
mg5_pulley_fixed.png
mg5_pulley_movable.png
mg5_rope_spool.png
char_director_idle.png
char_guard_sheet.png
ui_badge_momentum_crest.png
```

Keep originals in a `_raw/` folder if needed; **Godot should only import canonical names.**

15. ASSET CLEANUP PIPELINE

1. Generate or download.
2. Save raw (`_raw/`).
3. Rename to \$14.
4. Photopea: Magic Wand tolerance **25–40**, delete bg, Contract 1–2 px if halo.
5. Crop empty pixels; **keep integer pixel scale**.
6. Resize with **Nearest Neighbor** only.
7. Export **PNG**.
8. Checkerboard = transparency.
9. Compare next to player/factory floor.
10. Put in the correct folder (\$16).
11. Godot import: 2D, filter **Nearest**, no 3D compress.
12. AtlasTexture / SpriteFrames if a sheet.
13. Place in a **test scene** before replacing production art.
14. Log in `PROGRESS.md` + \$4 inventory.

Godot AtlasTexture example (School seated):

```
Rect2(1759, 848, 429, 607) on player_transparent.png (2816×1536).
```

16. GODOT FOLDER STRUCTURE

Current (CONFIRMED EXISTING): `res://asset/sprites/{character,Environment,animation,badge}` and `res://asset/music/`.

Proposed target (migrate later; do not break paths blindly):

```
asset/  
├─ characters/  
├─ environments/school|factory|powerplant|researchlab|observatory|industrial/  
├─ machinery/  
├─ minigames/mg1_emergency_brake/ ... mg5_pulley_rush/  
├─ ui/  
├─ vfx/  
├─ audio/{amb,music,sfx}/  
├─ fonts/  
├─ icons/  
└─ _raw/
```

Godot scenes stay under `final-godot/scenes/`.

Do not move files without updating every `res://` path.

17. AUDIO ASSET PLAN

Existing (DOWNLOADED, unwired)

Classroom ambience, opening sting, chair scrape, wood footsteps, paper rustle, school bell, typewriter tick.

Missing factory (P1 after MG1 exists)

Machinery hum, factory ambience, alarm, metal footsteps, stairs, door, quiz correct/incorrect, badge jingle, brake screech, gear clank, crane creak, hydraulic hiss, rope snap, motor whirl.

Zones 02–05

NOT YET DESIGNED in detail. Map B mentions per-floor loops (dock clank, drop-tower creak, shop grind, control hum). Record as **PLANNED** after zone lock.

Method: B (licensed SFX) is acceptable for mechanical sounds; C is for unique music stingers if needed.

18. UI MASTER PLAN

Component	Now	Need
World map	Painting + 1 factory hit area	Lock icons, 4 more zones
Dialogue box	DialoguePanel	Optional portraits/nameplate art
Quiz	QuizPanel MCQ	One-word input CONFIRMED IN DOCS , not in panel
Correct/incorrect	Text feedback	Icons optional
Progress / countdown	Missing	MG1/MG3
Badge collection	Flag only	Show <code>badge_momentum_crest</code>
Pause / retry / GAME OVER	Missing	MG fail path
PROBLEM SOLVED banner	Missing	MG1
Interaction icon	Missing	Optional P2
Godot theme / fonts	Default	P2

Programmer UI (Method E) is enough for defense MVP; custom art is P2.

19. CURRENT STATE SUMMARY (ONE PAGE)

What exists right now (2026-08-17)?

CONFIRMED / PLAYABLE

- Godot 4.7 GDScript project, autoloads, save, F5 reset
- School cutscene (teacher + seated student + dialogue + walk + fade)
- World map → factory exterior placeholder
- Floors 1–4: art, player, 80% MCQ, unlock gating
- Floor 1 collision relatively solid
- Dialogue + quiz UI
- Player movement on floors
- Headless `verify_project.gd`

DOWNLOADED BUT NOT INTEGRATED

- Director / manager / guard sheets
- True factory exterior PNG
- Badge PNG
- MG1–MG5 raw props (ramp, block, housing, gear, crane, press, rig, pulley, load, weights, gauge)
- Power plant & research lab select JPGs
- School audio pack

PARTIALLY READY

- Factory zone navigation and quizzes
- Exterior (wrong graphic, fake guard)
- NPC “system” (text only)

NOT READY

- Real five minigames + fail loop
- Guard/Director/Manager in scenes
- School→factory walk
- Factory audio
- Badge ceremony / Zone 02 unlock
- Zones 02–05 gameplay
- Custom fonts/icons
- Trained 50×4 question banks
- Scientist battle / flash-card combat

Rough Zone 01 completeness: quizzes/nav ~**70%**; story beats ~**40%**; minigames ~**10%**; audio integration ~**0%**.

20. ROADMAP AND DEPENDENCIES

This is **one** practical path, not the only one.

Phase	Focus	Depends on	Output
1	This master doc + PROGRESS.md	—	Shared truth
2	School + Factory navigation polish	Phase 1	Guard uses sheet; exterior PNG; optional schoolyard
3	NPCs on floors (manager/director sprites)	Cropped PNG sheets	Visible experts
4	Quiz content pass (still 5 Qs or expand)	JSON	Defense-ready questions
5	Mini-Game 1 real	MG1 art + UI	First real challenge
6–9	MG2–MG5	Each previous MG; shared fail→MG1	Emergency run
10	Crest + map unlock flag	MG5 + badge PNG	Zone 01 complete loop
11–14	Zones 02–05	Zone map decision + designs	New sites
15	Integration, audio, SUS polish	Playable loop	Capstone defense

Do not start Zone 02 implementation before §22 decisions.
Do not generate 40 decorative pipes before MG1 is playable.

21. PRIORITY SYSTEM

ID	Type	Task	Pri
P0-1	Design	Lock Zone 02–05 identity (Map A vs B)	P0
P0-2	Design	Lock pass threshold 80% vs 75%	P0
P0-3	Prog	Keep School as main scene	P0 (done)
P1-1	Asset+Prog	Wire Guard sheet; drop ColorRect	P1
P1-2	Asset+Prog	Use <code>factory_exterior</code> or confirm cutaway is intentional	P1
P1-3	Prog	Implement MG1 with existing ramp/block	P1
P1-4	Prog	GAME OVER → reload MG1	P1
P1-5	Prog	Director emergency before MG chain	P1
P1-6	Content	Replace placeholder quiz items	P1
P1-7	Asset D	Rename/clean JPG NPC sheets → PNG	P1
P2-1	Audio	Play school SFX in cutscene	P2
P2-2	Asset	Schoolyard + road signs	P2
P2-3	UI	Badge screen	P2
P2-4	Prog	Floor 2–4 ActionBar / collision polish	P2
P3-1	Content	50 questions × 4 floors	P3
P3-2	Design+Prog	Scientist battles	P3
P3-3	Asset C	Zone 02–05 worlds	P3

22. UNKNOWN / DECISION REQUIRED

Do **not** silently pick a side in code.

1. **Zone map A vs B** (Power Plant/Electricity vs Energy Refinery/Work-Energy, etc.).
 2. **Art: 2.5D vs 2D GDD lock.**
 3. **Language:** GDScript (now) vs C# (proposal).
 4. **Pass mark:** 80% everywhere in code vs 75% in one GDD quick-ref.
 5. **Factory height:** 5-story illustration vs 4 playable floors + rooftop.
 6. **Scene 2:** Player-controlled walk from school with road signs vs current cutscene→map→exterior.
 7. **Quiz mix:** MCQ-only vs MCQ + one-word (docs).
 8. **Floor 4 Continue** target: minigames vs more story.
 9. **World map:** only Factory clickable; when do other JPGs become buttons?
 10. **Bilingual Bengali+English fonts:** mentioned, not specified.
 11. **Sprite scale:** Player/School use **0.17**; future sheets must match.
 12. **Reference filename:** `factory_asset_reference.png` VS `physics_quest_factory_asset_reference.png`.
 13. **Scientist/flash-card battles:** in or out of MVP.
 14. **Mini-game fail:** docs = full MG1 restart; not coded.
 15. **Audio:** exist but not connected — bug or not yet scheduled?
-

23. PROGRESS.MD RULE

This master file is **baseline architecture**.

Daily/weekly status lives only in:

- `docs/PROGRESS.md`
- optional copy: repo-root `PROGRESS.md` if you keep one, **update both or pick one canonical** (`docs/PROGRESS.md` recommended)

`PROGRESS.md` must include: Current Status, Recently Completed, Currently Working On, Next Tasks, Blockers, Newly Added Assets/Scenes/Systems, Known Bugs, Decisions Made, Assets Still Missing, Next Recommended Action. **Use dates.**

24. AI DEVELOPER RULES

1. Never assume an asset exists unless it is in \$4 or you just added it.
 2. Never replace classroom/player/floor art without checking current scenes.
 3. Never invent a new visual style if `factory_asset_reference.png` or player/teacher sheets apply.
 4. Reuse: Floor 2 conveyor is already in the background PNG.
 5. Keep `snake_case` names; delete (2) duplicates.
 6. Keep pixel scale (Nearest, ~0.17 for characters).
 7. Gameplay-critical before decoration.
 8. Do not build Zones 02–05 before \$22.1.
 9. Label CONFIRMED vs PLANNED in every summary.
 10. If uncertain, ask; do not invent quiz science or zone stories.
 11. Update `PROGRESS.md` after meaningful work.
 12. New assets → inventory.
 13. New scenes → scene list.
 14. New systems → `PROGRESS.md`.
 15. Do not claim minigames are done because placeholder tscn files exist.
 16. Do not claim Guard is done because a PNG exists.
 17. Headless verify before calling a scene complete.
 18. F5 in-game resets save only; editor Play restarts the game.
-

25. QUICK START FOR A NEW AI

1. Open `final-godot/project.godot` — main scene School.
2. Play: School cutscene → Map → Exterior → Floor 1.
3. Read this file §19 and §22.
4. Read `docs/PROGRESS.md`.
5. Implement **one** P1 item (usually MG1 or Guard).
6. Do not start Zone 02.

Canonical paths

- Godot: `final-godot/`
 - Art/audio: `asset/`
 - Design dumps: `docs/capstone.md.txt`, `docs/physics_quest_zones_02-05_gdd (2).md`, `docs/physics_quest_zone02_scene_design (2).md`
-

26. EVIDENCE SOURCES AND HOW TO USE THEM

A new AI must treat these as **separate** sources. Never merge them into one “everything is done” story.

Source	Path / location	What it proves	What it does not prove
Godot project	final-godot/	Scenes, scripts, autoloads, main scene, pass threshold in code	That art on disk is used
Asset folder	asset/	Files downloaded/generated	That they are wired, cleaned, or final
Scene files	final-godot/scenes/**/*.*tscn	What is actually instantiated	Visual quality
Headless verify	scripts/tools/verify_project.gd	Structural checks the script knows about	Gameplay fun, art polish, audio
Chat/GDD dump	docs/capstone.md.txt	Zone 01 story, 5 minigames, asset wishlists	Implementation
Zones 02–05 GDD	docs/physics_quest_zones_02-05_gdd (2).md	Map B curriculum + Zone 02 full design	That Map B is the shipped map
Zone 02 scene design	docs/physics_quest_zone02_scene_design (2).md	Proposed Energy Refinery rooms	Built scenes
This master	docs/PHYSICS_QUEST_MASTER.md	Baseline inventory dated 2026-08-17	Live daily status
Living status	docs/PROGRESS.md	What changed after this baseline	Architecture

Proposal / capstone PDF (C#, 12-week methodology, SUS, Hake’s gain) is **CONFIRMED IN DOCUMENTATION** only. It is **not** in this repo as a built evaluation system.

27. SCREENSHOT / VISUAL CONFIRMATION NOTES

The following matches **what the design request asked to document** plus **what Godot scenes actually contain**. If a screenshot was provided in chat, it is corroborating evidence for art, not for systems that are not in the `.tscn`.

27.1 School / opening — VISIBLE IN GAMEPLAY

Status: CONFIRMED / VISIBLE

Evidence: `School.tscn` + `SchoolScene.gd` (and provided classroom screenshot if present in chat).

Visible / painted into `classroom.png` (not separate Godot nodes unless noted):

Element	How it exists
Classroom room	Full-viewport TextureRect <code>classroom.png</code>
Teacher	Separate sprite <code>teacher_idle.png</code>
Student desks	Painted in background
Teacher desk	Painted in background
Blackboard + physics formulas	Painted in background
Clock	Painted in background
Globe	Painted in background
Windows	Painted in background
Player / student	Sprite2D AtlasTexture on <code>player_transparent.png</code>
Scene transition	Automated fade via <code>SceneManager</code> after walk-to-door (legacy Next Scene hidden)

Cutscene behaviour **VISIBLE IN GAMEPLAY**: seated pose `Rect2(1759, 848, 429, 607)` → teacher line → student “Understood, Sir...” → idle front `Rect2(134, 117, 429, 587)` → tween left `Rect2(2283, 117, 429, 586)` → tween up to door `Rect2(1754, 117, 429, 587)` → World Map. Desk/door normalized positions: `DESK_NORM (0.47, 0.60)`, `AISLE_NORM (0.17, 0.60)`, `DOOR_NORM (0.12, 0.16)`.

27.2 Zone 01 — Factory Floor 1 — VISIBLE IN GAMEPLAY

Status: CONFIRMED / VISIBLE

Evidence: `Zone01_Floor1.tscn` + `factory_floor_1.png` (and provided Floor 1 screenshot if present).

Typical visible content of the floor painting (do not invent extra colliders): industrial floor, pipes, staircase area, factory lighting. Player WASD. Quiz/start interaction via dialogue + `QuizPanel1`. Transition toward Floor 2 after ≥80%. Collision: baked `wallColliders`; player layer 1, mask 2; walls layer 2.

27.3 Zone 01 — Factory Floor 2 — VISIBLE IN GAMEPLAY

Status: CONFIRMED / VISIBLE

Evidence: `Zone01_Floor2.tscn` + `factory_floor_2.png`.

Painted in the background (not separate animated nodes): conveyor language, robotic/industrial arms, machinery, pipes, crates, industrial equipment. Transition toward Floor 3 after quiz. **Do not** claim separate conveyor sprites exist just because they are painted.

27.4 What screenshots do not confirm

- Zone 02 Power Plant is **not** implemented. `powerplant_select.jpg` is a **select image on disk**, unused.
- Floors 3–4 art exists in Godot; if a screenshot was not provided, still treat them as **VISIBLE IN GAMEPLAY** because the scenes launch in the playable chain.
- Minigame screenshots of ColorRect placeholders do **not** mean the physics games exist.

28. GODOT SYSTEMS ARCHITECTURE

CONFIRMED EXISTING

Piece	Path	Role
Engine	Godot 4.7, Forward Plus, Jolt 3D (unused for this 2D game)	—
Main scene	res://scenes/School/School.tscn	Launch
GameManager	scripts/GameManager.gd	Save user://savegame.json , floor unlock flags, F5 reset
SceneManager	scripts/SceneManager.gd	Fade + change scene
BadgeManager	scripts/BadgeManager.gd	Badge flags (Momentum Crest flag can exist without UI)
Player	scenes/Player/Player.tscn	WASD on floors
School student	SchoolScene.gd Sprite2D	Cutscene only; physics off while seated
ZoneFloorController	scripts/ZoneFloorController.gd	Dialogue → quiz → continue
QuizManager	scripts/QuizManager.gd	Loads JSON; default pass 0.80
QuizPanel	scenes/UI/QuizPanel.tscn	MCQ UI
DialoguePanel	scenes/UI/DialoguePanel.tscn	One line + Continue
FloorRoomSetup	scripts/FloorRoomSetup.gd	Runtime walls on some floors
PlaceholderScene	minigame tscn scripts	Title + Next only
Verify	scripts/tools/verify_project.gd	Headless --script

Run headless verify (CONFIRMED EXISTING workflow):

```
Godot_v4.7.1-stable_win64_console.exe --path final-godot --headless --script res://scripts/tools/verify_project.gd
```

Godot exe used in this project’s notes: D:\Download\Godot_v4.7.1-stable_win64_console.exe — **UNKNOWN** if that path exists on every machine; do not invent another path.

Input: WASD + arrows (project.godot).

Stretch: canvas_items + expand.

F5: resets save flags only; **does not reload** the running scene.

Scene graph (playable):

```
School.tscn
→ WorldMap.tscn
  → Zone01_Exterior.tscn
    → Zone01_Floor1.tscn
      → Zone01_Floor2.tscn
        → Zone01_Floor3.tscn
          → Zone01_Floor4.tscn
            → MiniGame01_Brake.tscn (placeholder)
              → MiniGame02_Gear.tscn
                → MiniGame03_Balance.tscn
                  → MiniGame04_Pressure.tscn
                    → MiniGame05_Pulley.tscn
                      → WorldMap.tscn
```

29. QUESTION-BANK INVENTORY

CONFIRMED EXISTING under `final-godot/data/`:

File	Floor	Questions in file	Pass in JSON	Bloom tags present?
zone01_floor1_questions.json	1	5	0.80	Yes (Remember / Understand / Apply on items)
zone01_floor2_questions.json	2	5	0.80 (if same pattern)	Check file; do not assume 50
zone01_floor3_questions.json	3	5	0.80	Same
zone01_floor4_questions.json	4	5	0.80	Same

CONFIRMED IN DOCUMENTATION (not built): 50 questions per floor, 200 per zone, MCQ + one-word mix.

VISIBLE IN GAMEPLAY: `QuizPanel` is **MCQ only**. One-word input is **MISSING**.

Floor 1 sample (CONFIRMED EXISTING content, placeholder quality): SI unit of force; momentum = mv; Newton’s First Law; F=ma numerical; etc. These are **not** the trained Bloom bank from the proposal.

30. DUPLICATE-FILE AND IMPORT HYGIENE

Windows copy suffixes **CONFIRMED EXISTING** throughout `asset/` :

- `filename (2).png` / `.ogg` / `.jpg` next to the canonical file
- `filename.png (2).import` Godot import clones
- Spaces: `manager 1.jpg`
- Typos: `guard_shee`, `pully`, `guage`

Rule: Godot should reference **only** the name without `(2)` . `(2)` copies caused UID warnings. One duplicate `.tres` (`player_sprite_frames (2).tres`) was deleted; **many image/audio (2) files remain**.

Do not delete `(2)` files in bulk without checking Godot `.tscn` paths first. Prefer: stop importing them, then delete after a verify pass.

31. GAMEPLAY-CRITICAL VS DECORATIVE

Use this when choosing Nano Banana work. **P0/P1 before decoration.**

Gameplay-critical (Zone 01 MVP)

Asset / system	Why critical	Status
Player readable sprite	Movement identity	CONFIRMED EXISTING (wired)
Teacher idle	School beat	CONFIRMED EXISTING (wired)
Classroom, floors 1–4, world map	Navigation	CONFIRMED EXISTING (wired)
Quiz + dialogue UI	Assessment	CONFIRMED EXISTING (Godot UI)
Guard sprite in Exterior	GDD entrance beat	DOWNLOADED, unused
Manager / Director crops	Floor experts	DOWNLOADED JPG, unused
MG1 ramp + block	First real challenge	DOWNLOADED, unused in tscn
Fail → MG1 restart logic	Emergency run	MISSING (programming)
Momentum Crest display	Zone reward	PNG unused; UI MISSING
Pass-threshold consistency	Pedagogy	80% in code; 75% in one doc

Important but not blocking a first MG1 demo

True factory exterior, schoolyard, factory SFX, badge ceremony, Floor 2–4 collision polish, extra gears/pulleys.

Decorative / later

Extra workers, confetti, rooftop Floor 5, pipe variants, scientist battles, Zones 02–05 worlds, bilingual custom fonts.

Do not generate observatory telescopes before MG1 is playable.

32. FULL FACTORY SCENE SCRIPT (DOCS VS GODOT)

Authoritative story beats: docs/capstone,md.txt (user messages dated 2026-08-17). **CONFIRMED IN DOCUMENTATION.**

#	Documented beat	Godot 2026-08-17
1	Classroom: teacher assigns field work; back-bencher; leave classroom; black cutout	Partial: assignment is 80% line, not full "visit five sites / collect badges" speech. Leave classroom yes . Schoolyard no .
2	Appear in front of school; player-controlled walk to factory; road signs	MISSING — World Map substitutes
3	Guard greets (expected him; assignment not turned in); steps left ; door revealed; enter; black cutout	Partial: Exterior exists; Guard is ColorRect; no step-aside animation; door not a separate sprite
4	Floor 1: machinery; manager blocks stairs; quiz ≥80%; manager moves; stairs	Partial: quiz yes; manager sprite no ; stairs lock logic exists
5–6	Floors 2–3 same pattern, different machinery	Partial: same as Floor 1 (quiz, no manager sprite)
7	Floor 4: Director working, "tangled with work," quiz >80%, about to give badge, phone emergency , player offers help, minigames	Missing Director sprite + emergency cutscene. Continue goes to placeholder MG chain
8	Five minigames one run; any GAME OVER → MG1	Placeholder scenes only; fail loop not implemented
9	Momentum Crest; World Map unlocks Zone 02	Flag possible; no badge UI; no Zone 02

Contradiction in the same dump: one summary says Director receives the player on the **ground floor** and sends them to Floor 1; the scene script says Guard at the door and Manager on Floor 1, Director on Floor 4. **UNKNOWN / REQUIRES DECISION.** Do not invent a third version. Current Godot: Exterior Guard placeholder → Floor 1 quiz, no Director until Floor 4 text pattern.

33. FULL MINI-GAME DESIGN SPECS (AUTHORITATIVE GDD)

Source: docs/capstone,md.txt . Status of **all five** in Godot: **PLACEHOLDER ONLY**. Numbers below are **CONFIRMED IN DOCUMENTATION**, not implemented.

33.1 MG1 — Emergency Brake (Friction)

- Physics: ($f = \mu mg$), Newton's Second Law, deceleration
- View: side-view ramp; metal block sliding; speed indicator; red countdown (crash at zero)
- Data: mass **40 kg**; steel on concrete; ($\mu = 0.35$); distance **7 m**
- Step 1: Does friction alone stop the block? **YES / NO**. Correct = **NO**
- Step 2: extra braking force, 3 options (one correct)
- Success: decelerate, screech, dust, green **PROBLEM SOLVED**
- Fail: crash, sparks, alarm, GAME OVER + factory damage
- Assets: see §7; existing loose files `mg1_ramp_bg.png`, `mg1_block (1).png`

33.2 MG2 — Gear Swap (Gear Ratios)

- Physics: gear ratio = driven teeth ÷ driving teeth; output RPM
- Empty slot highlighted red; smoke from broken housing; RPM meter at 0
- Target RPM example: **150–200**; formula card: Output RPM = Input RPM × (input teeth ÷ output teeth)
- Four gears with tooth counts; drag into slot
- **Test before confirm** (one preview swap); confirm wrong → belt snap GAME OVER
- Existing: `mg2_housing.png`, `mg2_gear_a (1).png` (need more sizes)

33.3 MG3 — Counterweight Balance (Moments)

- Physics: ($F_1 d_1 = F_2 d_2$)
- Crane, conical load descending; left progress bar = timer
- Load **300 N** at **2 m**; CW point **3 m** right
- Weights: **150 / 180 / 200 / 220 / 250 N**
- ($F_2 = (F_1 \times d_1) / d_2$) → click block → **Place** (no preview)
- Timeout or wrong → crash GAME OVER
- Existing: `crane.png`, `load (1).png`, `weights (1).png` (may be a sheet — slice required)

33.4 MG4 — Pressure Panic (Pascal)

- Physics: ($P = F/A$), ($P_{\text{in}} = P_{\text{out}}$)
- Two pistons + pipe; safety valve needs exact pressure
- Gauge zones: green safe, yellow too low, red burst
- ($A_{\text{in}} = 0.02 \text{ m}^2$), ($A_{\text{out}} = 0.10 \text{ m}^2$), ($P_{\text{out}} = 5000 \text{ Pa}$)
- Force buttons **60 / 100 / 140 N**
- Slow dramatic needle; green → steam; else GAME OVER
- Existing: `press.png`, `guage (2).png`

33.5 MG5 — Pulley Rush (Mechanical Advantage)

- Physics: ($MA = \frac{\text{Load}}{\text{Effort}}$), Work = $F \times d$
- Empty rig; engine block; motor max **150 N**; load **600 N**; need **MA ≥ 4**
- Drag: single fixed, single movable, double movable, rope spool
- Live MA readout; rearrange free until **Engage Motor**
- Success: lift + Director clap; fail: rope snap
- Existing: `rig.png`, `pully (1).png`; motor / extra pulley types **MISSING** or unconfirmed as separate files

33.6 Emergency run rule

Any GAME OVER → restart **Mini-Game 1**. All five in one continuous run. **NOT IMPLEMENTED**.

34. COMPLETE NANO BANANA PROMPT CATALOG

34.1 Master template (copy exactly, then replace the last sentence)

Create a game-ready pixel-art asset for Physics Quest, a retro 2.5D educational physics RPG.

Use the uploaded `factory_asset_reference.png` as the primary visual style reference.

Match its pixel density, dark outlines, muted industrial purple-gray palette, shading, proportions, and mechanical design language.

Create only the requested object, centered, isolated, transparent background, no text, no UI, no extra objects, no background scene.

Do not use photorealism or 3D rendering. Pokémon Emerald-inspired readability. One asset only.

Then add the **subject sentence** from the tables below.

34.2 Factory / mechanical (upload `factory_asset_reference.png`)

Output filename	Subject sentence to append	Pri	Method if file already exists
<code>factory_conveyor_belt.png</code>	Create a side-view industrial conveyor belt segment with a flat top and visible rollers.	P1	Floor 2 already paints a belt — generate only if you need a separate moving sprite
<code>factory_conveyor_belt_broken.png</code>	Same belt with a snapped belt and a small puff of smoke, still a single isolated object.	P1	MG2
<code>factory_movable_crate.png</code>	Small wooden crate, three-quarter top-down, readable silhouette, no labels.	P2	Decorative / Map B crate game later
<code>factory_heavy_crate.png</code>	Larger darker crate with metal corners that reads as heavier.	P2	
<code>factory_pressure_plate.png</code>	Recessed metal floor pressure plate, top-down.	P2	
<code>factory_switch_industrial.png</code>	Chunky wall lever switch, isolated.	P2	
<code>factory_spring_mechanism.png</code>	Metal coil spring standing upright, isolated.	P2	
<code>factory_fulcrum.png</code>	Triangular fulcrum block for a lever, isolated.	P2	
<code>factory_gear_assembly.png</code>	Two meshed gears on a small mount, isolated.	P1	Prefer extra MG2 gears first
<code>factory_control_panel.png</code>	Chunky factory control panel with blank gauges and no readable numbers.	P2	
<code>factory_pipe_valve.png</code>	Short pipe segment with a wheel valve, isolated.	P3	Decorative
<code>factory_metal_platform.png</code>	Simple metal grated platform tile, isolated, no scene.	P3	
<code>mg2_gear_small.png</code>	Small industrial spur gear, few teeth, isolated, no numbers.	P1	Have <code>mg2_gear_a</code> — this is extra sizes
<code>mg2_gear_medium.png</code>	Medium spur gear, more teeth than small, isolated.	P1	
<code>mg2_gear_large.png</code>	Large spur gear, many teeth, isolated.	P1	
<code>mg2_empty_slot.png</code>	Empty circular gear housing hole, slight red highlight baked in, isolated.	P1	Or Godot ColorRect
<code>mg3_weight_150n.png</code>	Single rectangular weight block, isolated, no text (we add 150 N in Godot).	P1	Slice <code>weights (1).png</code> first (Method D)
<code>mg4_steam_puff.png</code>	Small steam puff VFX, isolated, transparent.	P2	
<code>mg4_pipe_burst.png</code>	Burst pipe with water/steam spray, isolated.	P2	
<code>mg5_pulley_fixed.png</code>	Single fixed pulley wheel with a bracket, isolated, no rope text.	P1	Cleanup <code>pully (1).png</code> first
<code>mg5_pulley_movable.png</code>	Single movable pulley block, isolated.	P1	
<code>mg5_pulley_double.png</code>	Double movable pulley block with two wheels, isolated.	P1	
<code>mg5_rope_spool.png</code>	Industrial rope spool, isolated.	P1	
<code>mg5_motor.png</code>	Small industrial motor, isolated.	P1	
<code>mg1_sparks.png</code>	Spark burst VFX for a crash, isolated.	P2	
<code>mg1_dust.png</code>	Dust puff VFX for a screech stop, isolated.	P2	

Output filename	Subject sentence to append	Pri	Method if file already exists
ui_banner_problem_solved.png	Optional pixel banner shape with no text (text in Godot).	P2	Method E preferred
npc_factory_worker.png	See character prompts below.	P3	

34.3 Character prompts

Guard — Method D first on `guard_shee (1).png`. Only if unusable:

Upload: `player_transparent.png` + `factory_asset_reference.png`

Create a game-ready pixel-art factory security guard for Physics Quest.
Match the uploaded player sheet’s body proportions, pixel density, dark outlines, and top-down / three-quarter camera.
Idle front pose only, factory uniform, isolated, transparent background, no pose labels, no photorealism, no text.

Worker — same uploads as Guard:

Create a game-ready pixel-art factory worker for Physics Quest.
Match the uploaded player sheet’s proportions and pixel density.
Idle front pose only, industrial coveralls, isolated, transparent background, no text, no extra tools unless a single wrench in hand.

Director / Manager — do **not** regenerate until Method D (crop existing JPG sheets) is tried. Existing files: `director_sheet.jpg`, `manager 1.jpg`, `manager 2.jpg`, `manager 3.jpg`.

34.4 Other-zone prompts (P3 — zone map unlocked first)

Upload `powerplant_select.jpg` + `factory_asset_reference.png`:

Create a game-ready pixel-art electrical generator, isolated, transparent background, no text, matching the uploaded industrial pixel style. Physics Quest asset.

Create a game-ready pixel-art industrial turbine, isolated, transparent background, no text, matching the uploaded industrial pixel style.

Upload `researchlab_select.jpg`:

Create a game-ready pixel-art laboratory electromagnet coil, isolated, transparent background, no text, matching the uploaded scientific pixel style.

Create a game-ready pixel-art horseshoe or lab magnet, isolated, transparent background, no text.

Observatory / thermodynamics: **do not mass-generate**. No on-disk style target. Mark **NOT YET DESIGNED**.

35. ZONE 02–05 DOCUMENTED DESIGNS (MAP B)

These are **CONFIRMED IN DOCUMENTATION** in `physics_quest_zones_02-05_gdd (2).md` and the Zone 02 scene-design file. They are **not** Godot scenes. Map A names (Power Plant / Spark Emblem, etc.) remain **CONFIRMED IN DOCUMENTATION** from the original chat summary. **Do not build until §22.1 is decided.**

35.1 Shared Map B skeleton (docs)

Each zone: 4 floors, Bloom tiers, **50 questions/floor**, one signature mini-game per floor, capstone on top. This **conflicts** with Zone 01's "quizzes on floors + five emergency minigames after Floor 4." **UNKNOWN** whether Zones 02–05 copy Zone 01's emergency-run or Map B's per-floor minigame.

35.2 Zone 02 Energy Refinery (full design in docs)

NPC: **Foreman Rina**. Theme: potential → kinetic; fault is inefficiency.

Floor	Topic	Mini-game (docs)	Proposed rooms (scene design doc)
Hub	Entry	—	Exterior / hub, amber pipes, Rina at gate
1	Work & Power $W=Fd$, $P=W/t$	Loading Dock crate push + timer	Wide dock, force gauge, distance ticks
2	KE/PE conservation	Drop Tower + pendulum	Tall rail, height markers
3	Simple machines, MA	Machine Shop benches	Lever / pulley / ramp benches
4	Efficiency capstone	Fix the Refinery chained system	Control room, schematic of floors 1–3

Assets listed in that GDD (spec only): Rina sheet, workers, crates, force meter, drop tower, pendulum, modular machines, per-floor audio loops.

On disk today: `powerplant_select.jpg` **only** (Map A flavor). **No** Rina sprite. **No** refinery floors.

35.3 Zone 03 Matter Labs (skeleton)

Floors: states of matter sort; $P=F/A$ gauges; density ID; buoyancy vessel design; capstone sinking submarine. Disk: `researchlab_select.jpg` unused.

35.4 Zone 04 Wave Observatory (skeleton)

Floors: thermal expansion; heat transfer routing; wave generator; echo sonar; capstone resonance door. Disk: **nothing found**.

35.5 Zone 05 Light & Circuits Citadel (skeleton, provisional)

Floors: mirrors; lenses; Ohm circuits; electromagnet lift. Docs flag possible split **05a/05b**. Disk: **nothing found**.

35.6 Map A proposed rooms (NOT FINAL)

If Map A wins, **PROPOSED / TO BE DESIGNED** (user roadmap, not a locked GDD):

- Power Plant: exterior, entrance, generator, turbine, control, electrical challenge, quiz, final, Spark Emblem
- Research Lab: magnetism challenge, Flux Badge
- Observatory: optics, Lens Token
- Industrial: thermodynamics, Heat Sigil

Do not treat that list as Floor 1–4 locked design.

36. UI COMPONENT MASTER LIST

Component	Required by	Now	Method
World map	Navigation	Painting + 1 hotspot	A + E
Zone markers (5)	Map	1 factory only	E / C
Locked zone icon	Map	Missing	E or icon PNG
Dialogue box	Story	DialoguePanel	E done
Character nameplate	Story	Partial / text in line	E
Question box	Quiz	QuizPanel	E done
MCQ buttons	Quiz	QuizPanel	E done
One-word answer input	GDD	Missing	E
Submit button	Quiz	Present for MCQ	E
Correct / incorrect feedback	Quiz	Text	E; icons P2
Explanation panel	Quiz	In panel	E
Progress bar	MG1/MG3	Missing	E
Countdown	MG1	Missing	E
Badge display	Zone clear	PNG unused	E + A
Inventory / badge collection	Meta	Missing	E
Pause menu	Polish	Missing	E
Retry button	MG fail	Missing	E
Continue button	Floors	Present	E
Interaction icon	Optional	Missing	C or E
Success banner PROBLEM SOLVED	MG1	Missing	E
Failure banner GAME OVER	MG	Missing	E
Godot default theme	All UI	In use	—
Custom fonts	Bilingual?	Folder empty	B + UNKNOWN

37. AUDIO REQUIREMENT MATRICES

37.1 School pack — DOWNLOADED, not played

File	Suggested use	Wired?
amb_classroom.ogg	School loop	No
music_opening_sting.ogg	After teacher line / fade	No
sfx_chair_scrape.ogg	Stand up from desk	No
sfx_footsteps_wood.ogg	Walk to door	No
sfx_paper_rustle.ogg	Assignment beat	No
sfx_school_bell.ogg	Scene start	No
sfx_typewriter_tick.ogg	Dialogue optional	No

Duplicates (2) exist for each; use canonical names only.

37.2 Factory / MG — MISSING (P1 after MG1 exists)

Cue	Suggested filename	Pri
Machinery hum loop	amb_factory.ogg	P1
Factory ambience	amb_factory_floor.ogg	P1
Alarm / siren	sfx_alarm.ogg	P1
Metal footsteps	sfx_footsteps_metal.ogg	P2
Stair climb	sfx_stairs.ogg	P2
Door / guard move	sfx_door_metal.ogg	P2
Quiz correct	sfx_quiz_correct.ogg	P1
Quiz incorrect	sfx_quiz_wrong.ogg	P1
Badge reward	sfx_badge_jingle.ogg	P2
Brake screech	sfx_mg1_screech.ogg	P1
Crash	sfx_mg1_crash.ogg	P1
Gear clank	sfx_mg2_clank.ogg	P1
Belt snap	sfx_mg2_snap.ogg	P1
Crane creak	sfx_mg3_creak.ogg	P1
Impact	sfx_mg3_crash.ogg	P1
Hydraulic hiss	sfx_mg4_hiss.ogg	P1
Pipe burst	sfx_mg4_burst.ogg	P1
Motor whirl	sfx_mg5_motor.ogg	P1
Rope snap	sfx_mg5_snap.ogg	P1
Voice blip	sfx_dialogue_blip.ogg	P3

Method B (licensed SFX) is preferred over generating audio in image tools.

37.3 Zones 02–05

Map B mentions dock clank, drop-tower creak, shop grind, control hum. **PLANNED** after zone lock. No files.

38. CHARACTER POSE AND ANIMATION MATRIX

Character	Idle front	Side	Back	Walk	Talk	Special	On disk	In scene
Player	Yes (sheet)	Yes	Yes	Partial SpriteFrames	Distinct talk unconfirmed	Seated crop used in School; climb stairs MISSING	player_transparent	School + floors
Teacher	Idle crop	Sheet unused	—	—	—	—	teacher_idle wired; full sheets unused	School
Guard	Sheet likely	Sheet likely	—	—	—	Side-step MISSING as anim	guard_shee unused	ColorRect
Manager 1–3	JPG sheets	JPG	—	—	—	Shift off stairs MISSING	Unused JPG	Text only
Director	JPG	JPG	—	—	—	Worried / clap unconfirmed as separate files	Unused JPG	Text only
Workers	—	—	—	—	—	Working pose	MISSING	—
Foreman Rina	—	—	—	—	—	Map B only	MISSING	—

Player scale in School/floors notes: **~0.17**. Match when cropping NPC idle frames (~429×587 teacher idle as a size hint, not a law).

39. GODOT IMPORT AND SPRITE-SCALE RULES

1. Import as **2D**.
 2. Filter: **Nearest**.
 3. No lossy 3D compression on pixel art.
 4. Resize outside Godot with **nearest neighbor** only.
 5. AtlasTexture crops must be integer pixels. Known School crops are documented in §3.1 / §15.
 6. Do not mix 3D-rendered assets into factory floors.
 7. After import, instance in a **sandbox scene** before replacing production `School1.tscn` / floor scenes.
 8. Update this inventory and `PROGRESS.md`.
-

40. PHOTOPEA CLEANUP PROCEDURE (STEP-BY-STEP)

CONFIRMED IN DOCUMENTATION (chat dump). Use for JPG sheets and white-background PNG.

1. Open photopea.com → File → Open.
2. Magic Wand (W), Tolerance **25–40** (start ~30).
3. Click white/light background; Shift-click remaining islands.
4. Inspect marching ants — if they eat the shirt, lower tolerance; if background remains, raise 5–10.
5. Delete. Checkerboard = transparency.
6. Select → Modify → Contract 1–2 px → Delete again to kill halo.
7. Crop empty margin; keep integer size.
8. Image Size only if needed; **constrain proportions**; interpolation **Nearest Neighbor**.
9. File → Export as → **PNG**.
10. Name with \$14 (`guard_sheet.png` , not `guard_shee (1).png`).
11. For multi-pose sheets, **leave pose labels** during BG removal; crop labels out in Godot AtlasTexture.
12. Put file in the correct `asset/` folder; Godot import \$39.
13. Test in-scene.

Do not use blurry bicubic upscale. **Do not** export JPG for characters.

41. WHAT TO IMPLEMENT NEXT (RECIPES)

These are **next actions**, not claims of completion. After each, update `PROGRESS.md`.

41.1 P1 — Wire Guard (Asset D + Prog)

1. Confirm `guard_shee (1).png` is usable.
2. Rename copy to `char_guard_sheet.png` (keep original until verify).
3. Photopea transparency if needed.
4. AtlasTexture idle front.
5. Replace Exterior ColorRect.
6. Optional: tween left for door reveal (GDD).
7. Headless verify + play Exterior.

41.2 P1 — Exterior art decision (Design + Asset)

1. Compare `factory_exterior (1).png` vs current `factory.png` cutaway.
2. Pick one; document in `PROGRESS.md` Decisions.
3. Do not use both as competing full-screen BGs.

41.3 P1 — Mini-Game 1 (Prog + existing art)

1. Open `MiniGame01_Brake.tscn`; stop using ColorRect-only placeholder.
2. TextureRect `mg1_ramp_bg.png`; Sprite `mg1_block (1).png`.
3. Godot UI: data panel (40 kg, μ 0.35, 7 m), YES/NO, three force buttons, countdown.
4. Implement correct NO → force select → stop vs crash.
5. GAME OVER → reload MG1 (zone-run rule can wait until MG2 exists, but document it).
6. Do not wait for custom banners; Labels are enough.

41.4 P1 — NPC sheets on floors (Asset D + Prog)

Crop manager 1–3 and director from JPG → PNG → Sprite2D on Floors 1–4. Dialogue already exists.

41.5 P1 — Director emergency (Content + Prog)

Script the phone beat from §32 before MG1. No new zone required.

41.6 Do not do yet

- Zone 02 scenes
 - 50×4 question generation as a blocker
 - Scientist battles
 - Regenerating classroom/player
 - Mass Nano Banana of observatory/thermo props
-

42. SCIENTIST / FLASH-CARD BATTLE (SIDE CONCEPT ONLY)

CONFIRMED IN DOCUMENTATION (late chat in `capstone.md.txt`): Pokémon-like scientist pick, HP, flash cards that unlock attacks via questions, per-zone battles, side content beside the main assignment.

Status: PLANNED / OPTIONAL / NOT IN CODE. Conflicts with current manager-quiz floors. **Do not implement** unless the team explicitly schedules it after Zone 01 MVP.

APPENDIX A — CANONICAL PATHS

Item	Path
Repo root	C:\Users\Mufrid Johanee\Desktop\capstone\final\
Godot project	final-godot/
Assets	asset/ → res://asset/
This document	docs/PHYSICS_QUEST_MASTER.md
Living status	docs/PROGRESS.md
Zone 01 GDD dump	docs/capstone,md.txt
Map B GDD	docs/physics_quest_zones_02-05_gdd (2).md
Zone 02 rooms	docs/physics_quest_zone02_scene_design (2).md
Style reference	asset/sprites/Environment/factory_asset_reference.png
Save file	user://savegame.json

APPENDIX B — STATUS LEGEND (REPEAT)

Label	Meaning
CONFIRMED EXISTING	On disk in this repo
CONFIRMED IN DOCUMENTATION	Specified in GDD/proposal/chat
VISIBLE IN GAMEPLAY	Reachable in Godot
DOWNLOADED / AVAILABLE	File exists; may be unused
GENERATED BUT NEEDS CLEANUP	AI/export; JPG/typo/white bg/unwired
PLANNED	Intended
MISSING	Needed; no usable file
NOT YET DESIGNED	Named only
UNKNOWN / REQUIRES DECISION	Do not invent

End of master document. Evidence dated 2026-08-17. If the repo changes, update §4 and `PROGRESS.md`, not vibes. Regenerate the PDF after large inventory edits.